Southern Luzon State University

COLLEGE OF ENGINEERING

Lucban, Quezon

Exploratory Data Analysis Output

Geospatial Machine Learning-Based Identification of High-Risk Fire Zones in San Francisco for Emergency Response Optimization

In Partial Fulfillment of the requirements for

CPE 15L Cognate/Professional Course Laboratory

Prepared by:

Tenedero, Sean Carr Geoffrey M.

BSCPE 3A – IE

Presented to:

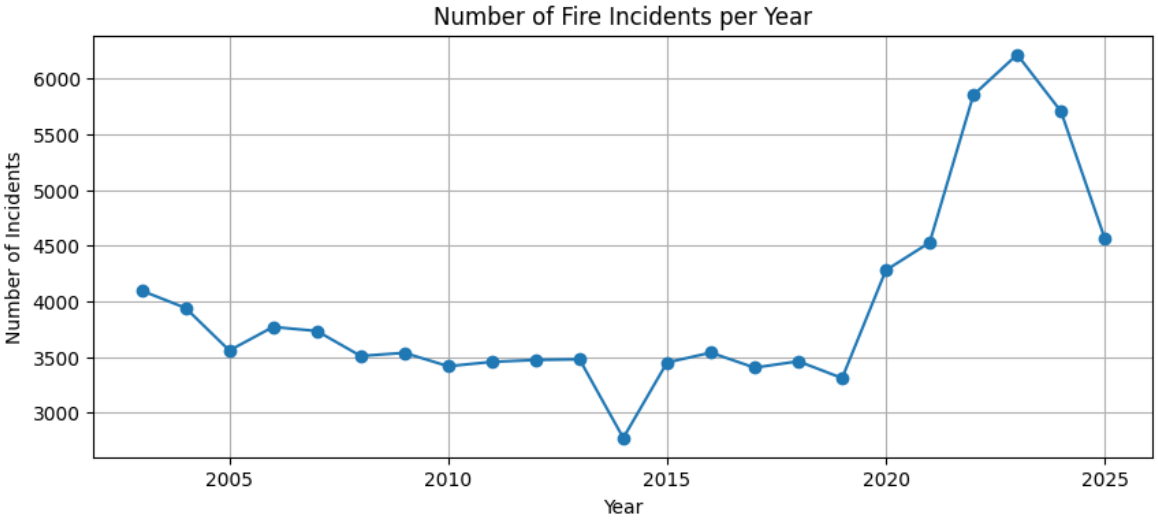
Engr. Julie Ann Susa-Gili

Date Submitted:

December 12, 2025

YEARLY FIRE INCIDENT TRENDS

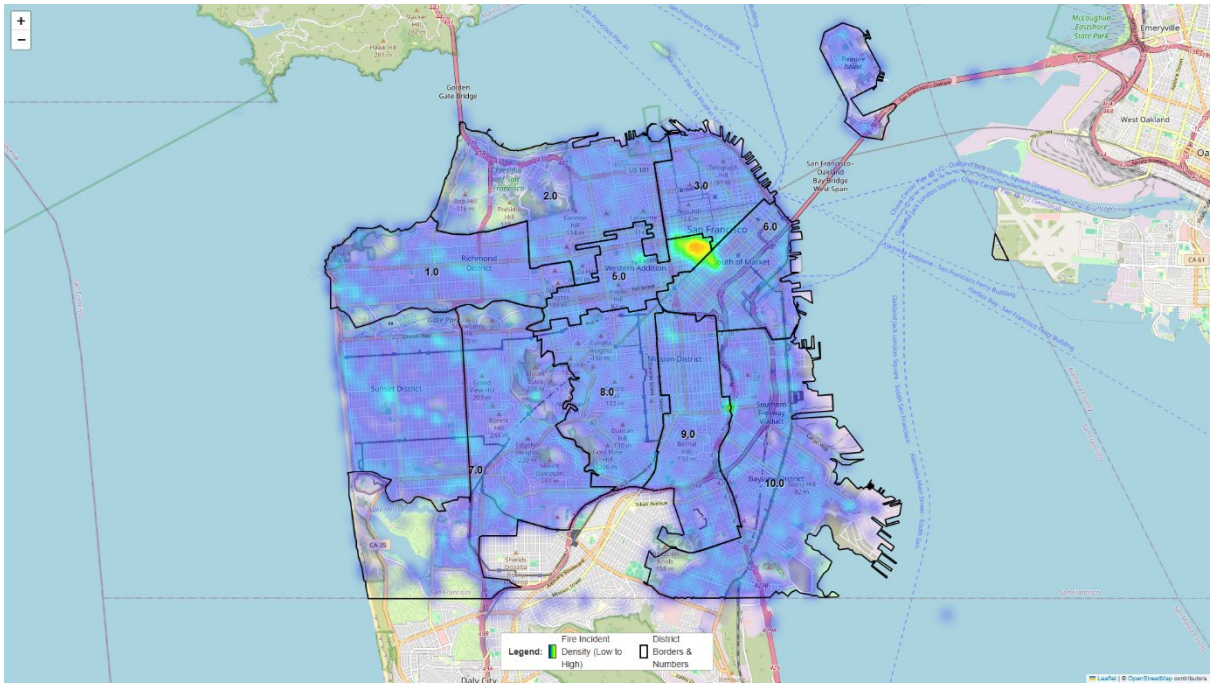
This visualization shows the number of fire incidents recorded in San Francisco for each year in the dataset. By plotting these counts, we can observe temporal patterns, identify years with unusually high or low fire activity, and assess long-term trends. Understanding how fire incidents fluctuate over time helps inform resource planning, policy decisions, and further analysis of factors influencing fire risk.



The period from 2019 onward shows a marked escalation in fire risk, possibly due to environmental, social, or reporting changes. The overall pattern suggests a stable baseline with recent volatility.

AGGREGATE AND MAP FIRE INCIDENTS BY SUPERVISOR DISTRICT

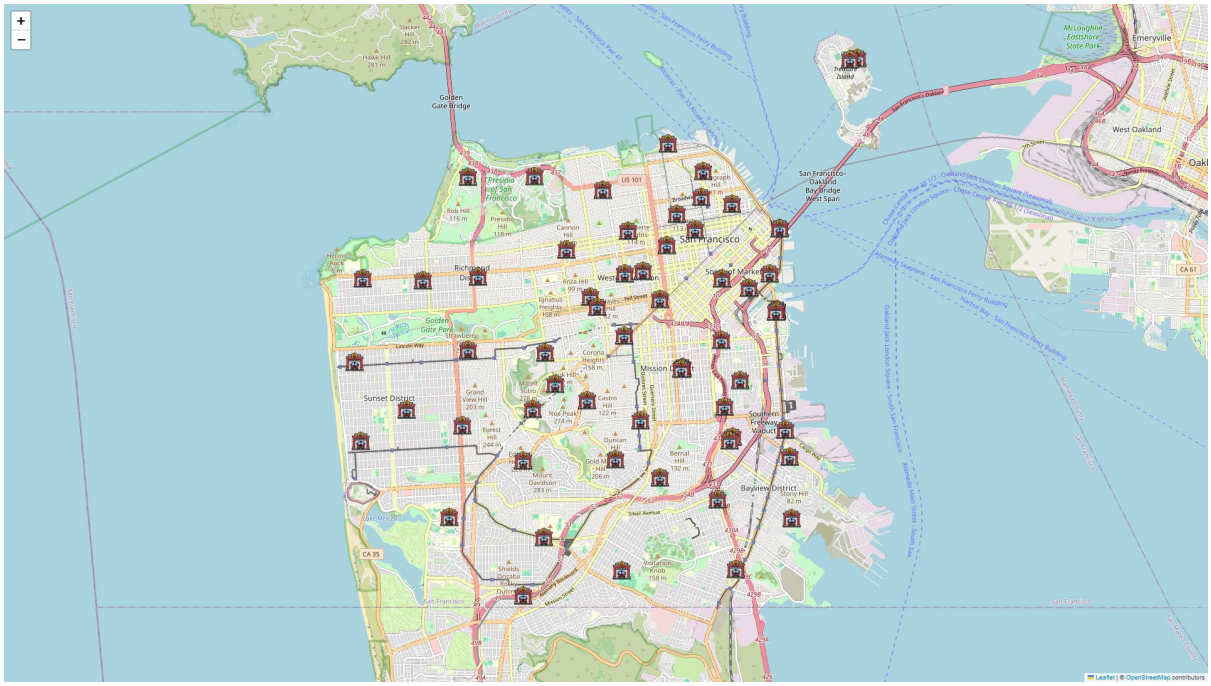
This cell groups fire incidents by supervisor district, merges the incident counts with the district boundaries, filters out invalid or overly large districts, and visualizes the results as a choropleth map. The map shows the spatial distribution of fire incidents across San Francisco, helping to identify districts with higher fire risk.



The heatmap visualization shows that fire incidents in San Francisco are highly concentrated in the downtown and northeast districts, with the most intense hotspot centered around the urban core. In contrast, outlying districts such as Sunset and Richmond experience significantly lower incident densities. This spatial pattern suggests that fire risk is closely linked to areas with higher population density and commercial activity, highlighting the need for focused emergency response and resource allocation in the city’s central neighborhoods.

INTERACTIVE MAP OF SAN FRANCISCO FIRE STATIONS

This interactive map displays the locations of all San Francisco fire department facilities using custom fire station icons. Each marker provides a popup with the station's name and address. The map is centered on San Francisco and helps visualize the spatial distribution of emergency response resources across the city. Use this map to explore where fire stations are located relative to neighborhoods and incident hotspots.



The map of fire station locations across San Francisco shows a well-balanced distribution, with stations positioned throughout all major districts. This even coverage ensures that emergency response resources are accessible citywide, supporting rapid deployment and effective service to both central and outlying neighborhoods.